BELL STATE SIMULATOR

DAY 22 | 30-DAY QUANTUM COMPUTING CHALLENGE

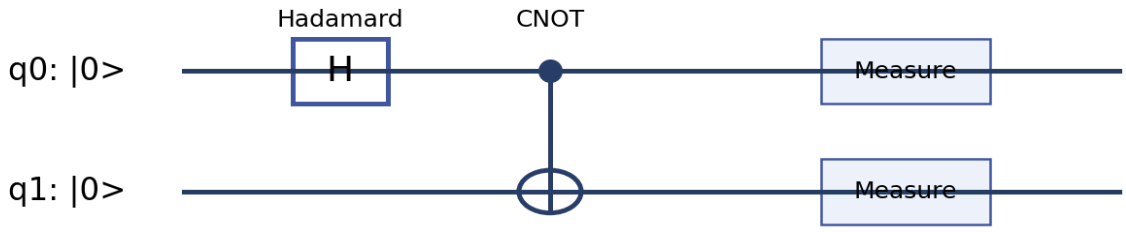
Experiment goal

Prepare a two-qubit Bell state, measure it 1,000 times, and visualize why the results are correlated.

1. Why this project matters

Unlike two independent coin flips, Bell-state measurements have a joint pattern: in an ideal computational-basis measurement, both qubits return the same bit. This is a first practical introduction to entanglement.

2. Circuit and gate sequence



Start in |00>. Apply Hadamard (H) to qubit 0, then CNOT with qubit 0 as control and qubit 1 as target. Measure both qubits.

3. State transformation

Initial: |00> → after H: (|00> + |10>)/sqrt(2) → after CNOT: (|00> + |11>)/sqrt(2).

The final state is the Bell state Phi-plus. The squared amplitudes give P(00)=0.5, P(11)=0.5, P(01)=0 and P(10)=0.

4. Expected results

Outcome	Theoretical probability	Expected count (1,000 shots)
00	50%	About 500
01	0%	0
10	0%	0
11	50%	About 500

EXPERIMENT & EXECUTION

5. Tools and environment

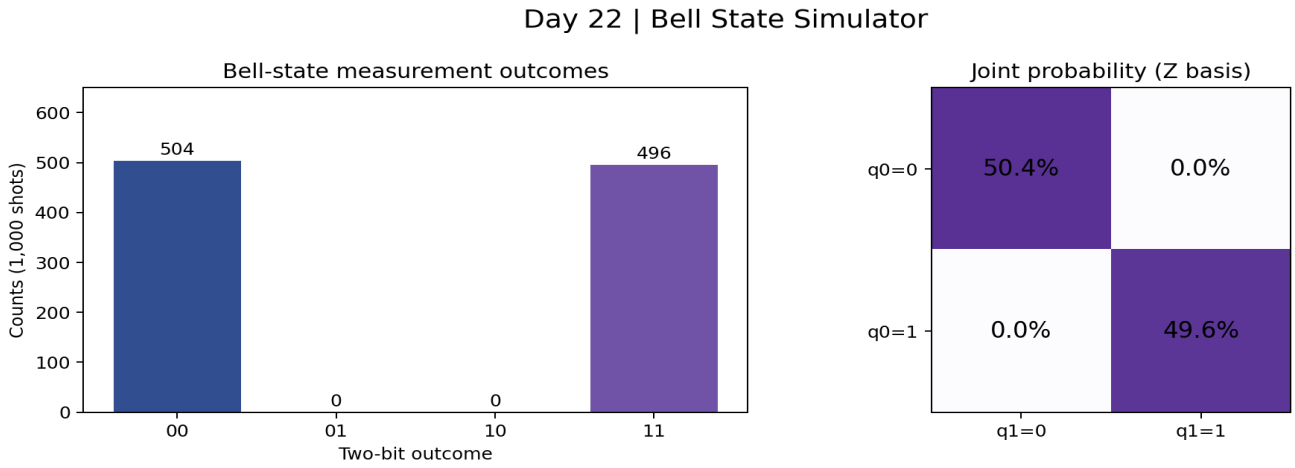
Python 3, NumPy, Matplotlib, and Jupyter Notebook. Optional: Qiskit and Qiskit Aer to reproduce the circuit with a gate-based simulator.

6. Run the notebook

1. Open **bell_state_simulator.ipynb** in Jupyter or VS Code.
2. Install dependencies: `pip install numpy matplotlib notebook`.
3. Run the cells from top to bottom.
4. Inspect the statevector, count table, bar chart and probability heatmap.
5. Optionally enable the Qiskit section after installing Qiskit Aer.

7. Recorded results (seed=22, 1,000 shots)

The reproducible ideal simulation generated: 00: 504, 01: 0, 10: 0, 11: 496. The exact 00/11 split changes with a different random seed; 01 and 10 remain impossible in this ideal model.



8. How to interpret the plots

The bar chart shows two populated outcomes, 00 and 11. The joint-probability heatmap has nonzero values only along its diagonal. Both individual qubits are approximately 50/50 random, while the pair is perfectly correlated in the Z basis.

Important distinction

A matching 00/11 pattern alone does not establish entanglement experimentally; classical correlated bits can show the same pattern. Entanglement certification needs additional measurement settings or a suitable witness.

REVIEW & EXTENSION

9. Everyday analogy

Imagine two sealed envelopes that always contain matching symbols. Opening one tells you what the other contains. The analogy explains correlation, but not the uniquely quantum nature of entanglement.

10. Common mistakes

• Assuming the result must be exactly 500/500 in every 1,000-shot run.
• Confusing correlation in one measurement basis with proof of entanglement.
• Claiming entanglement can send messages faster than light.
• Treating a noiseless simulator as equivalent to physical quantum hardware.

11. Try these experiments

• Change the shot count to 100 and 10,000.
• Change the random seed and compare the 00/11 split.
• Add an X gate on the second qubit after Bell-state preparation: predict the new allowed outcomes.
• Run the same circuit in Qiskit Aer and compare the counts.

12. Summary: What to remember

H on qubit 0 followed by CNOT creates the Bell state (|00> + |11>)/sqrt(2). In an ideal Z-basis measurement, only 00 and 11 appear, each with probability 1/2. A histogram demonstrates the predicted correlation; deeper tests are required to verify entanglement.

Project files

bell_state_simulator.ipynb — executable experiment
results.png — generated results visualization
circuit.png — gate circuit illustration
README.md — setup and GitHub overview